High Validity, Low Transfer: Anchored Evaluation of Diffusion Counterfactual Explanations for Chest Radiographs

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Abstract

Counterfactual explanations for image classifiers are usually judged by validity: how often an edit flips the decision of the classifier that guided it. Because the same model steers and scores the edit, high validity cannot distinguish a class-relevant change from an adversarial perturbation. We propose an anchored evaluation protocol for chest radiograph counterfactuals that pairs validity with two measurements, each calibrated against explicit nulls: a held-out transfer ratio, which re-scores counterfactuals with classifiers never used for guidance and is bracketed by an adversarial-attack floor and a real-image ceiling, and an in-lung Lesion Hit Rate against radiologist-drawn opacity boxes. We apply it on RSNA Pneumonia to a PPO-scheduled latent diffusion generator (PPO-CF) and to an iterative Latent Shift baseline. Normal-to-pneumonia counterfactuals reach 94.5% validity, yet three held-out classifiers reproduce only 28% of the guidance shift, no more than a norm-matched attack on the guidance models; Latent Shift behaves the same way. Pneumonia-to-normal edits, the direction with annotated lesions, transfer well above the attack floor and do target lesions, but weakly (1.16× chance within the lung, about half of it explained by where lesions typically lie). A whole-image null would score them 2.6×, below the 3.2× of an edit spread uniformly over the lung. A fixed schedule at the policy's mean action matches the learned policy in validity and lesion targeting, so the controller contributes an aggressive schedule rather than per-image adaptation. Validity should be reported only with such anchored, null-calibrated checks.

1. Introduction

Classifiers deployed on chest radiographs are typically audited with attribution maps, which indicate where evidence is concentrated but not what change would alter the prediction. Visual counterfactual explanations close this gap: given a query image x with prediction $f(x)$ and a target class c, a generator produces x_c that (i) is scored at c by the classifier, (ii) remains a realistic radiograph, and (iii) differs from x only in a minimal, class-relevant region, preserving anatomy, foreign objects and patient identity.

The standard success criterion, Counterfactual Validity (CV), is almost always computed with the same classifier whose gradients guided generation. It therefore cannot distinguish a class-relevant edit from an adversarial perturbation tuned to one model. This paper measures how much that matters for one system. We pair CV with two companion measurements, one anchored to human annotation (a Lesion Hit Rate against radiologist-drawn opacity boxes) and one to model independence (a held-out transfer ratio computed with classifiers that never guided generation), and apply them to a diffusion counterfactual generator we built. The concern is not hypothetical there: our system reaches 94.5% CV in the normal-to-pneumonia (n→p; reverse p→n) direction while the held-out classifiers reproduce on average 28% of the guidance shift (Fig. 1a), no more than a norm-matched attack on the guidance models. An iterative Latent Shift [5] baseline run under the same protocol shows the same pattern. Two systems do not establish how common this is, and we claim only that nothing in the conventional metric suite would have surfaced it for either. Both companion metrics need their own nulls: scored against the whole image, a lesion-blind edit that merely stays in the lung would beat every method we test (§4.2).

The generator, PPO-CF, is the test case. It frames per-step guidance control as a sequential decision problem: a PPO agent sets the sampling schedule (strength, guidance scales, pass count, stopping) of a frozen, LoRA-adapted latent diffusion model, on the hypothesis that one hand-tuned schedule fits a heterogeneous test population poorly; FiRe [31] instead hand-designs adaptive guidance and early stopping (§2). The agent never touches the diffusion weights. The evaluation also tests that hypothesis, and refutes it. A fixed schedule set to the policy's mean ac-

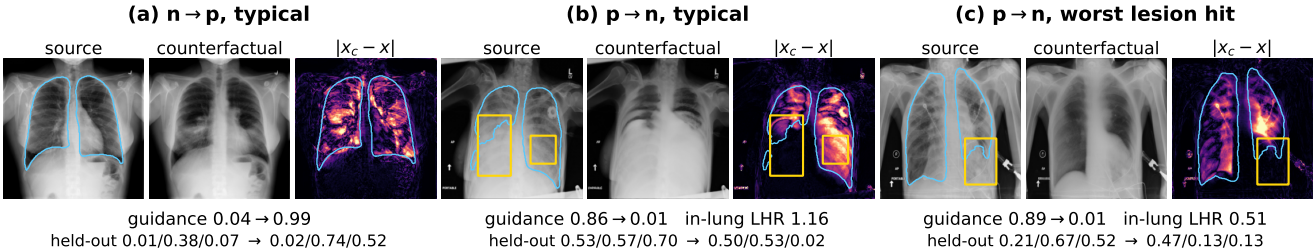


Figure 1. Rule-chosen valid edits of the default policy at (a) the median held-out shift (n→p), (b) the median and (c) the lowest in-lung LHR (p→n). Blue: lung mask; yellow: opacity boxes; $|x_c - x|$ scaled per image. Scores: guidance ensemble; held-out `nih`/`pc`/`mimic_ch`.

tion matches the policy in lesion targeting and CV (§5.2), so the protocol, not the controller, is the contribution we stand behind.

Contributions.

1. An evaluation protocol that pairs CV with an in-lung lesion hit rate, calibrated against lung-uniform, mismatched-box and autoencoder nulls, and a direction-specific held-out transfer ratio, bracketed by a matched attack floor and a real-image ceiling, together with the split-, seeding- and routing-level controls an arm comparison needs in order to mean anything (§4).
2. PPO-CF, a PPO-scheduled sampling controller around a frozen, LoRA-adapted Stable Diffusion generator with a soft, segmentation-gated background lock (§3), used as the case study.
3. Findings (§5): high CV with n→p transfer at the attack floor, for our system and for Latent Shift; weak but real lesion targeting, matched by a fixed schedule; a whole-image null that inflates lesion targeting about sevenfold and creates arm rankings that vanish within the lung; and two localization mechanisms that, as implemented, do not help.

2. Related Work

Counterfactuals for chest radiographs. Singla *et al*. [25] introduced a GAN-based counterfactual generator for chest radiographs with the evaluation protocol (CV, FID and Foreign-Object Preservation, FOP) that became the de facto benchmark, which we adopt for comparability; FOP needs a detector trained on private annotations, so §4.1 uses a model-free substitute. Latent Shift [5] moves an autoencoder latent along the classifier gradient, RadEdit [17] edits radiographs with a diffusion model under multiple masks to stress-test classifiers, and FastDiME [30] accelerates diffusion counterfactuals with self-optimized masks, including on chest X-ray shortcuts. Diffusion counterfactuals for natural images [1, 11, 12] and in latent space (LDCE [8]) likewise score validity with the target classifier.

Evaluating beyond the guidance classifier. Re-scoring with a model not used for generation has precedents: DiME and ACE [11, 12] count non-target attribute changes with an independently trained oracle, and DVCE [1] uses adversarially robust classifiers so that gradient-guided edits become semantic. Our transfer ratio re-scores the *target* shift itself, as a per-direction ratio to the guidance shift on validators trained on other corpora, read against a matched attack floor and a real-image ceiling rather than a cut-off; the novelty is the calibrated pairing with CV and with a lesion anchor, not re-scoring as such. LHR adapts the energy-based pointing game of Score-CAM [27] (the share of a map's energy inside a box) from saliency maps to the counterfactual difference map, with an in-lung area null that a lesion-blind edit cannot beat (§4.2).

Diffusion editing and spatial locks. Classifier-guided diffusion [7, 16, 18] has displaced GANs for editable generation. We build on a LoRA-adapted [9] Stable Diffusion v1.5 [19] backbone with DDIM sampling [26]. RePaint [14] and Blended Latent Diffusion [2] re-composite the unedited region at every step; we adapt this as a soft lock through a dilated, feathered lung mask.

Localized and adaptive counterfactuals. ECED [15] edits a Score-CAM foreground and DocVCE [22] filters adversarial gradients and restores unneeded patches. FiRe [31] is closest in mechanism: it refines at a fixed noise level under saliency dual masks and *hand-designs* the controls PPO-CF learns, guidance that strengthens when more editing is needed and weakens near the target decision, and early stopping once the decision flips. PPO-CF instead learns per-step strength, guidance scales, pass count and stopping from a reward, around a lung rather than a saliency mask (a Grad-CAM mask collapsed our task, §5.5). Our finding that a fixed schedule at the policy's mean action recovers most of its gain (§5.2) suggests a hand-designed controller of FiRe's kind may suffice. None of these three methods reports a lesion-anchored metric or held-out target transfer, and none was run on our split and seeding, so we do not compare head-to-head (Supp. E3).

RL and schedules for diffusion. Most RL work on diffusion fine-tunes the weights against a reward (*e.g.* DDPO [3]). Sampling schedules have been optimized without RL, for noise levels (Align Your Steps [21]) and for guidance weights over time [29], where simple monotone schedules are hard to beat. Learning per-step sampling and guidance controls with RL around a frozen generator, for counterfactual editing, is less explored; in our setting the learned controls turn out close to a fixed schedule.

3. Method

3.1. Problem formulation

Given a frozen binary classifier $f : \mathcal{X} \to [0, 1]$ and a target class $c \in \{0, 1\}$, we seek x_c that trades off two objectives,

$$\min_{x_c} \Big(\underbrace{\mathrm{BCE}(f(x_c), c)}_{\text{classification}}, \underbrace{d(x, x_c)}_{\text{preservation}} \Big), \tag{1}$$

which we do not combine with a single fixed weight. The trade-off is set during sampling by the ratio of the classifier- to preservation-guidance scales, which the policy chooses per pass (§3.2, §3.3), and during policy training by the reward weights (Supp. Table R1). Here d is a masked distance penalizing change inside the segmented lung field and change to a frozen object-detector embedding (a foreign-object proxy), and leaving change outside the lung field free, subject to the soft background lock below, which largely removes that freedom away from the lung field.

3.2. Latent generator and background lock

Images are encoded to a $4 \times 32 \times 32$ latent by a frozen SD1.5 VAE. A lightweight MLP maps $(c, \text{pooled latent})$ to a single cross-attention conditioning token, replacing SD1.5's text conditioning with a scalar class signal. Rank-8 LoRA adapters are inserted into the UNet's attention projections and trained on the standard ϵ-prediction objective on the RSNA training fold (§4.4), adapting the natural-image prior toward radiography without touching the VAE or the bulk of the UNet.

Guided DDIM sampling. Generation runs a partial, SDEdit-style forward-then-reverse DDIM trajectory whose length is a strength fraction of $T_{\text{diff}} = 50$ steps. At each step the UNet's noise prediction gives both the predicted clean latent $\hat{z}_0$ and the DDIM update z_{t-1}. Two guidance gradients are taken with respect to $\hat{z}_0$, through the VAE decoder D, and added to z_{t-1}:

$$z_{t-1} \leftarrow z_{t-1} - w(i)\, m \odot \big(\alpha_{\text{cls}} \nabla_{\hat{z}_0} \mathcal{L}_{\text{cls}} + \alpha_{\text{pres}} \nabla_{\hat{z}_0} \mathcal{L}_{\text{pres}}\big), \tag{2}$$

where $\mathcal{L}_{\text{cls}} = \mathrm{BCE}(f(D(\hat{z}_0)), c)$, $\mathcal{L}_{\text{pres}} = \mathcal{L}_1^{\text{seg}} + \eta\, \mathrm{MSE}_{\text{obj}}$, $\mathcal{L}_1^{\text{seg}}$ is the lung-mask-weighted L_1 distance from the original image, $\mathrm{MSE}_{\text{obj}}$ compares object-detector features, and $\alpha_{\text{cls}}, \alpha_{\text{pres}}$ include the per-pass scales set by the policy (values in Supp. Table R4). The update is gated by the soft edit mask m defined below and by a linear ramp $w(i)$ rising from 0 at a learned fraction of the trajectory, which avoids the discontinuity of a hard guidance switch.

Background lock. After each guided step the latent is blended toward an anchor: the original image's latent forward-noised to the next timestep, and used clean at the final step,

$$z_{t-1} \leftarrow m \odot z_{t-1} + (1 - m) \odot \tilde{z}_{t-1}^{\text{orig}}. \tag{3}$$

Here $m \in [0, 1]$ is the soft edit mask: the original image's lung-probability map, resized to the 32×32 latent grid, dilated by two latent cells (≈ 16 px) and Gaussian-feathered (σ of one cell). It is not thresholded. The blend does not depend on how the guidance gradient behaves: where $m \approx 0$, far from the lungs, the final latent is essentially the original image's, rather than being repainted by the free SDEdit process, which is what an unconstrained high-strength schedule does to the entire thorax. The constraint is soft rather than exact. The dilated, feathered band around the lungs remains editable, and because the VAE decoder mixes neighbouring latent cells, preservation in pixel space outside the mask is approximate.

3.3. PPO-scheduled sampling

Rather than fixing strength, guidance scales and pass count, a PPO agent [23] emits a 9-dimensional continuous action at each of up to $T = 25$ policy steps, mapped through sigmoid transforms to the controls in Table 1. Each policy step runs two to four chained generator passes (two, plus one for each of a_4, a_5 thresholded at $\sigma(a) > 0.5$, identically in training and evaluation), with the exact mapping and per-pass schedule modifiers given in Supp. R2.

Observation and architecture. The observation is a flat vector of $8{,}192 + 65{,}536 + 3 = 73{,}731$ entries: the VAE latents of the original and current images (two $4 \times 32 \times 32$ latents, 8,192 entries; drawn from the VAE posterior rather than taken at its mean, with the current-image latent redrawn at every step, so the observation is stochastic even for a fixed image), the soft lung-segmentation mask of the current image at the 256×256 image resolution (65,536 entries, i.e. 88.9% of the vector), and three scalars: the step fraction t/T, the current distance $d_t = |f(x_t) - c|$ and the target c. The agent is a policy network with a shared trunk and policy, value and three auxiliary heads (feasibility, progress, stop); the trunk consumes the full flattened observation, and the policy head outputs nine Gaussian means with a state-independent, per-dimension learnable log-std. It is trained with the clipped PPO objective and GAE (hyperparameters

Table 1. Action space, emitted at every policy step from the image state and step index. a_0 is the base strength s; pass k applies $\min(0.9,\ s\beta(1{+}0.15k))$, β from a_6 (Supp. R2). In evaluation the n→p policy holds a_0, a_4, a_5, a_7 at or near their bounds (a_7's upper bound may bind) and the p→n policy varies all nine (Supp. R3). †Narrowed from $[0.5, 3.0]$ in an earlier code version (Supp. Table R5).

	Controls	Range
a_0	DDIM strength	$[0.20, 0.70]$
a_1	classifier-guidance scale	$[0.3, 1.2]^\dagger$
a_2	preservation-guidance scale	$[0.3, 1.5]$
a_3	stop signal τ_t	$[0, 1]$
a_4, a_5	extra pass, thresholded at $\sigma(a) > 0.5$	$\{0, 1\}$
a_6	per-pass strength boost	$[0.85, 1.15]$
a_7	guidance bias (cls vs. pres)	$[0.70, 1.30]$
a_8	guidance ramp-up start fraction	$[0.1, 0.5]$

in Supp. Table R3); episodes end when $d_t < \epsilon_{\text{stop}} = 0.08$ or after $T = 25$ steps.

Reward. The per-step reward is a weighted sum of: progress $d_t - d_{t+1}$; two distance penalties, $-d_{t+1}$ and $-d_{t+1}^2$; an L_1 fidelity penalty against the original image; a lung-mask-weighted L_1 preservation penalty plus a KL term between the foreign-object detector's softmaxed features on the original and current images; a constant per-step cost; a validity bonus when $d_{t+1} < \epsilon_{\text{stop}}$ (which also ends the episode); and a stop bonus when, in addition, $\tau_t > 0.5$. The sparsity arm adds an annealed L_1 term. The whole n→p reward is multiplied by 2.25. Weights and the implemented form of each term are listed in Supp. Table R1; Supp. R collects every setting needed to reproduce the method.

3.4. Directional policies

The environment is built around a fixed target, so one policy is trained for n→p ($c = 1$) and another for p→n ($c = 0$). Each direction fine-tunes its own LoRA adapter from the same data, seed and recipe, and the resulting weights differ (Supp. R4). The default and Grad-CAM arms are evaluated with each direction's own adapter. Through a loading default, the single-classifier, 3-model and sparsity arms used the n→p adapter for both directions, so where they are compared with the default we use the default evaluated under that same protocol (Table 4); its p→n LHR is then 0.028 higher (95% CI $[-0.000, +0.058]$; Supp. R4).

4. Evaluation Protocol

4.1. Conventional metrics

Following [25], the evaluation set is split by classifier confidence into $X_n = \{x : f(x) < 0.2\}$ and $X_p = \{x : f(x) > 0.8\}$, up to $n_{\text{eval}} = 200$ each, and each directional policy generates from the corresponding group with its deterministic (mean) action. Full definitions of every metric below are given in Supp. R6.

Counterfactual Validity. With $s = 2c - 1$ ($+1$ for n→p, -1 for p→n),

$$\text{CV} = \frac{1}{N}\sum_{i=1}^{N} \mathbf{1}[\, s\,\Delta_i \geq \tau \,\wedge\, B_i\,], \tag{4}$$

where $\Delta_i = f(x_c^i) - f(x^i)$, B_i requires $s\,(f(x_c^i) - 0.5) > 0$ and $s\,(f(x^i) - 0.5) \leq 0$, f is the guidance classifier (the ensemble mean in ensemble arms) and $\tau = 0.8$. On X_n and X_p the crossing condition B_i is redundant: $f(x) < 0.2$ and $s\,\Delta_i \geq 0.8$ already give $f(x_c) \geq 0.8$ for n→p, and symmetrically for p→n. The tightening over the original, direction-agnostic definition therefore comes from the sign requirement alone: an image moving the wrong way by a large margin is no longer credited.

FID. Between each direction's counterfactuals and the real images of its target group (X_p for n→p, X_n for p→n; selected by f, not by label), on 1024-d features of a TorchXRayVision DenseNet-121 [6] with `rsna` weights rather than Inception-v3. These weights also guide generation in every trained arm (the 3-model ensemble is `rsna`, `chex`, `mimic_nb`), so FID is not independent of guidance. With $n \leq 200$ samples in a 1024-d feature space FID is high-variance; we report its bootstrap interval and KID in Supp. E6 and do not rank arms by it.

Identity Preservation (IP). As a dependency-free substitute for FOP we report MS-SSIM and HF-corr $= (1 + r)/2$, where r is the Pearson correlation between the high-pass-filtered source and counterfactual (0.5 means uncorrelated), a proxy for whether thin structures such as tubes, leads and pacemakers survive editing.

Edit concentration. The share of $\sum |x_c - x|$ held by the top 5% of pixels, averaged over pairs; spatially uniform change scores exactly 0.05. We also report the fraction of pixels with $|x_c - x| > 0.01$; part of it is VAE reconstruction error, separated out in Supp. E6. There is no calibrated threshold; as an illustrative heuristic only, a compact lesion-shaped edit would place roughly half or more of its mass in the top 5%.

4.2. Ground-truth-anchored Lesion Hit Rate

CV, FID and IP share a blind spot: none of them can tell whether the edit landed on the pathology. RSNA ships 9,555 radiologist-drawn opacity boxes [24], which we retain. LHR adapts the energy-based pointing game [27] to

edits. With $\delta = |x_c - x|$, box mask A and lung mask L (lung probability > 0.5, the same segmenter that drives the lock), the per-image lift is

$$\ell = \frac{\sum_{A \cap L} \delta \,/\, \sum_L \delta}{|A \cap L| \,/\, |L|}, \tag{5}$$

and LHR is its mean over annotated images. The edit mass and the null are both restricted to the lung, so any lesion-blind edit that stays in the lung scores 1. LHR is defined only for p→n, since n→p sources carry no annotation (Supp. R6).

The restriction is not cosmetic. With the whole image as the null ($\sum_A \delta / \sum \delta$ over $|A|/HW$, the pointing-game convention), a uniform wash scores 1.00, but a uniform wash over the lung scores 3.21, because boxes lie in the lung and the lock keeps most edit mass there. Under that null, our default policy scores 2.60, below a lesion-blind in-lung edit. We report it only as context. Three further nulls calibrate Eq. (5) (Supp. E7). The VAE round trip $|D(E(x)) - x|$ alone scores 0.98, and measuring the edit against $D(E(x))$ instead of x leaves LHR unchanged, so autoencoder error does not create or hide lift. Scoring each edit against *other* patients' boxes gives the share of lift explained by where lesions typically lie, rather than by this image's lesion.

4.3. Held-out transfer

Because CV is computed with the guidance classifier, it cannot separate a class-relevant edit from an adversarial perturbation. We therefore re-score every counterfactual with three TorchXRayVision DenseNet-121 classifiers that never participated in guidance and report, per validator v and direction, the ratio of the validator's mean signed score shift to the guidance model's own shift over the same N images,

$$\rho_v = \frac{\sum_i s\left(f_v(x_c^i) - f_v(x^i)\right)}{\sum_i s\left(f(x_c^i) - f(x^i)\right)}, \tag{6}$$

a ratio of means rather than a mean of per-image ratios, undefined when the guidance shift is zero. 95% intervals are percentile bootstraps over the N images (10,000 resamples, numerator and denominator resampled jointly). The validators differ in how independent they are of RSNA. `pc` (PadChest [4]) and `mimic_ch` (MIMIC-CXR [13]) share no images with it. `nih` was trained on NIH ChestX-ray14 [28], of which RSNA's images are a subset [24], so it has probably seen our test fold's source images, though with NLP-mined labels rather than RSNA's radiologist labels, and its Pneumonia label is rare (≈1% prevalence). We therefore lead with the three-validator mean and report `nih` separately as the cell with the lowest transfer, not as the most independent one. Of the guidance models, `rsna` was trained on RSNA data and `chex` on CheXpert [10]. The denominator must be the shift of the classifier(s) that actually guided generation, so it is read from the configuration stamp each evaluation shard carries, which records the guidance classifiers for the default and later arms (the single-classifier arm's shards predate this key and are treated as single-classifier guidance). We do not threshold ρ_v. Two anchors calibrate it instead, both computed with the same machinery (Supp. E8). The *ceiling* is the ratio for real image pairs, the validator's mean positive-minus-negative score gap on the 5,337-image test fold over the guidance ensemble's gap. It is 0.47, 0.58 and 0.42 for `nih`, `pc` and `mimic_ch`, so even real disease does not transfer fully. The *floor* is the ratio of attacks on the guidance ensemble alone, applied to the same source images: FGSM and 10-step PGD with an L_∞ budget, and L_2 PGD with each image's budget set to its counterfactual's $\|x_c - x\|_2$. We also report decision flips at each validator's own Youden-optimal threshold on the test fold. Pneumonia is rare in `nih`'s training labels and RSNA's positive class is "Lung Opacity", so we repeat everything with the validators' Consolidation, Infiltration and Lung Opacity outputs; conclusions do not change (Supp. E8).

4.4. Controls the comparison requires

A control-arm comparison needs three preconditions that we had assumed and that did not hold (full account in Supp. E1): (i) one deterministic, persisted, patient-level split shared by LoRA fine-tuning, the PPO pool and evaluation, replacing a shared dataset object under which evaluation images had been seen during fine-tuning; (ii) seeded training, since two identical unseeded submissions produced two different policies; and (iii) control arms exercised end to end, which surfaced a routing bug and a shared output directory that would each have produced a well-formed but wrong table. Every reported run uses the 80/20 patient-level split with seed 0, LoRA seed 0, PPO seeds 1 (n→p) and 2 (p→n), and evaluation base seed 0; the seed replicates of Supp. E10 change the PPO seed, and one of them also the LoRA seed. Generator fine-tuning, PPO rollouts and the validity probe use only the 80% training fold; every evaluated image is in the 20% test fold. Every default-arm number in the paper comes from one evaluation in which each direction uses its own adapter, on the same 400 test images and per-image seeds as the no-policy, fixed-schedule and seed-replicate runs of §5.2 and §5.4. The single-classifier arm shares split, seeds, reward weights, action ranges and anchoring with the default arm and differs only in the guidance flag, and through it in the classifier-selected groups (78 of 200 p→n sources shared) and in hardware.

5. Experiments

5.1. Setup

We use the RSNA Pneumonia Detection Challenge dataset [20] (DICOM, min–max normalized, resized to

Table 2. Default configuration, $n = 200$ per group, from one evaluation in which each direction uses its own generator adapter (§4.4). LHR (Eq. 5) is defined only for p→n; the whole-image lift is context, not a test (§4.2).

Metric		n→p	p→n
CV	fraction valid, τ=0.8	0.945	0.670
	mean	0.808	
LHR	in-lung lift (chance = 1)	—	1.165
	$t = 7.73, p = 1.6 \times 10^{-12}, n = 148$		
	whole image (lung-uniform: 3.21)	—	2.595
Transfer	mean over 3 validators	0.281	0.276
EditConc.	top-5% mass (uniform=0.05)	0.254	0.314
	fraction of pixels changed	0.647	0.634
FID	vs. real target class	22.38	4.76
IP	MS-SSIM	0.924	0.868
	HF-corr	0.948	0.913

256×256, binary label collapsed from bounding-box annotations). In the default configuration the guidance classifier is the mean-probability ensemble of two TorchXRayVision DenseNet-121 models (`densenet121-res224-rsna`, `densenet121-res224-chex`); the single-classifier arm uses `rsna` alone; segmentation uses TorchXRayVision PSPNet for lung and heart channels; a YOLOv8n feature extractor supplies the foreign-object embedding. Each policy is trained for 800 episodes per direction; for the 2-model ensemble run the two directions took 8h40m (n→p, on an A100) and 11h00m (p→n, on an H100), both including LoRA fine-tuning (5.7 h and 5.1 h).

5.2. Main results

Table 2 reports the default configuration (2-model guidance ensemble, no Grad-CAM mask, no sparsity reward). Within the lung, p→n edits put 1.165× the box's area share of their mass on it (95% CI $[1.123, 1.207]$): real, but weak lesion targeting. About half of it is generic. Scored against other patients' boxes, the same edits reach 1.089, which leaves an image-specific part of $+0.076$ ($[+0.037, +0.117]$, $p = 3 \times 10^{-4}$). Over the generator's default schedule without a policy, run on the same images with the same guidance, groups, adapters and seeds, LHR rises from 1.088 (paired $+0.077$, $[+0.037, +0.120]$, $p = 5 \times 10^{-4}$, $d = 0.29$) and p→n CV from 0.225 to 0.670 (90 images gain validity, 1 loses it; McNemar $p = 7.4 \times 10^{-26}$). Under the whole-image null the LHR gap would read $+0.532$, seven times larger, mostly because the policy keeps more of its edit in the lung (66% vs. 56%). In n→p the policy adds nothing measurable (CV 0.945 vs. 0.940, 4 vs. 3 discordant). The edits are diffuse: a quarter to a third of the difference mass lies in the top 5% of pixels (uniform: 0.05), and about half the changed pixels are VAE round-trip error (Supp. E6).

Table 3. Held-out transfer ratio ρ_v (Pneumonia outputs) with its anchors, on the default's 200 + 200 source images unless marked. Ceiling: real positive vs. negative test images. Attacks target the guidance ensemble only; both reach guidance CV ≥ 0.96. *Own classifier-selected groups.

	nih		pc		mimic_ch	
	n→p	p→n	n→p	p→n	n→p	p→n
Real pairs (ceiling)	0.47		0.58		0.42	
Default policy	**0.04**	**0.17**	**0.47**	**0.31**	**0.33**	**0.36**
Single-classifier*	0.08	0.10	0.41	0.22	0.20	0.11
Latent Shift (iter.)	0.04	0.19	0.34	0.24	0.08	0.24
PGD L_∞, 4/255	0.22	0.02	0.31	0.02	0.23	0.07
PGD L_2, norm-matched	0.18	0.07	0.45	0.46	0.49	0.02

A fixed schedule matches the policy. To ask whether the gain comes from per-image adaptation, we hold every control at the policy's mean action over its evaluation steps, with the same step budget and validity-based stopping and no training. On the same images this fixed schedule is indistinguishable from the policy in LHR (1.162; paired $+0.003$, $[-0.030, +0.040]$, $p = 0.86$) and in p→n CV (0.660 vs. 0.670, 14 vs. 16 discordant), and in n→p it is valid on exactly the same 189 images. Its mean p→n transfer is lower (0.244 vs. 0.276). A fixed schedule picked from a grid by the training reward, without RL (Supp. E5), also matches the policy in LHR (1.177; -0.012, $[-0.047, +0.023]$) and p→n CV (0.720 vs. 0.670, $p = 0.12$), with lower transfer (0.211). Its whole-image LHR deficit (-0.187, significant) was a locality effect. What the policy contributed is an aggressive schedule, not per-image adaptation.

5.3. CV overstates what the edits achieve

Table 3 places the counterfactuals between their anchors. In p→n they sit well above the floor: an L_∞ attack that reaches guidance CV 0.96 transfers at 0.02–0.07, the policy at 0.17–0.36, about 60% of the real-pair ceiling. In n→p they do not. An L_2 attack with each counterfactual's own norm transfers as well or better on `pc` and `mimic_ch`, a $4/255$ L_∞ attack with a quarter of that norm reaches 0.22–0.31, and on `nih` the counterfactuals transfer less than every attack. Decision flips at the validators' own operating points agree: `pc`, `mimic_ch` and `nih` flip 91%, 69% and 22% of n→p counterfactuals, and 97%, 98% and 36% of norm-matched attacks. Transfer of 0.28 therefore does not mean that validators reject most edits. It means that the edits move held-out scores no more than direct optimization against the guidance models does. The norm-matched attack is large (L_∞ 0.37) and may itself add opacity-like structure, so this is a statement about what transfer can certify, not proof that the edits are adversarial.

This is the paper's central empirical claim. In n→p the default policy reaches a CV of 0.945, yet its held-out trans-

Table 4. Five trained arms. Bold marks the default. LHR is in-lung (whole-image in parentheses). Transfer is the mean over {nih, pc, mimic_ch}; *over {nih, pc} only (leak-free set). $n = 200$ per group, except the 3-model arm's p→n group ($n = 32$, 27 annotated). Paired in-lung LHR tests: §single-classifier policy rerun on the default's images and seeds; ‡against the default evaluated with the n→p adapter for both directions, as those arms were; Grad-CAM against the per-direction evaluation.

Arm	LHR	Transfer n→p/p→n	LHR vs. default (paired)
Single-classifier	1.084 (2.29)	0.230/0.145	worse (p=2×10^{-4})§
2-ens (default)	**1.165 (2.60)**	**0.281/0.276**	—
3-model ensemble	1.216 (2.49)	0.350/0.276*	unresolved (n=12)‡
Sparsity reward	1.167 (2.40)	0.311/0.212	n.s. (p=0.085)‡
Grad-CAM constraint	1.140 (1.64)	0.161/0.103	n.s. (p=0.37)

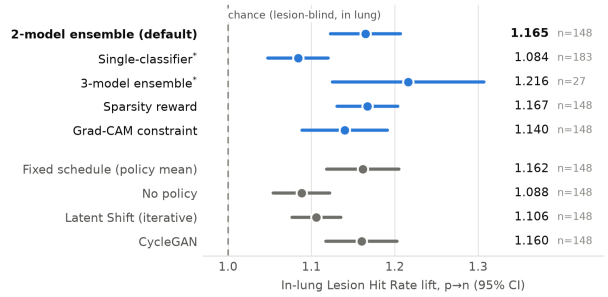


Figure 2. In-lung LHR (p→n, 95% CI) of the arms in Table 4 and four reference runs. 1.0: lesion-blind in-lung edit. *Own image groups.

fer gives no evidence of class-relevant content beyond an attack's. Iterative Latent Shift [5], a different generator (an autoencoder latent walk) under the same guidance, groups and stopping rule, shows the same pattern: CV 1.00 with mean n→p transfer 0.15 (Supp. E3). Nothing in the conventional metric suite surfaces this. Against single-classifier guidance, every ensemble-guided run across PPO seeds (§5.4) has lower p→n CV (0.495–0.765 vs. 0.800–0.850) but higher in-lung LHR (1.14–1.19 vs. 1.08–1.11) and p→n transfer (0.21–0.28 vs. 0.12–0.14); n→p transfer does not separate them. The metric that fell is the one that shares a classifier with the generator.

Fig. 1 shows what averages hide. In (a) the guidance score goes from 0.04 to 0.99 while `nih` barely moves and `pc` and `mimic_ch` move less than half as far. In (c) the edit is valid but puts only half the box's area share of its in-lung mass on the lesion (lift 0.51), and `nih` moves toward pneumonia.

5.4. Five trained arms

Every arm in Table 4 is a policy trained for 800 episodes per direction and evaluated end to end, with the split and seeds of §4.4. LHR verdicts are paired tests on shared images, never differences of group means. Under the in-lung null only one ranking survives, ensemble over single-classifier guidance, and it also survives PPO seed variation. Every other ranking that the whole-image lift produces (sparsity worse, Grad-CAM much worse, trained over random actions) disappears once edits are compared within the lung (Fig. 2).

Ensemble vs. single-classifier guidance, and seed variation. On the default's images and seeds with its own adapters, the single-classifier policy scores 1.083, no better than no policy (1.088); the ensemble's advantage, $+0.082$ ($[+0.042, +0.124]$), also holds on the 76 annotated sources the original groups share under matched adapters ($+0.074$, $p = 0.006$). New PPO seeds around the fixed adapters, plus an earlier full retrain of the default (LoRA seed 1, one shared adapter), give three runs per configuration (Supp. E10). Every ensemble run scores above every single-classifier run in in-lung LHR (8 of 9 pairs at $p < 0.01$; seed-averaged $+0.075$, $[+0.050, +0.102]$) and above no policy; no single-classifier seed does. p→n CV moves by up to 0.27 between ensemble seeds, so CV-based arm comparisons need more than one run per arm. Growing the ensemble to three members improves `pc` n→p transfer from 0.465 to 0.623 with non-overlapping CIs, does not resolve on LHR ($n = 12$), and lowers mean CV to 0.624.

A trained policy does not beat a random one in the lung. On its 183 annotated images, the single-classifier policy and a random-action policy with the same bounds score 1.084 and 1.081 (paired $+0.003$, $[-0.024, +0.031]$, $p = 0.85$). Their whole-image gap ($+0.090$, $p = 0.012$) is a locality effect.

5.5. Two localization mechanisms, as implemented

Both results below are diagnostics of our implementations, not evidence against sparsity or attribution priors in general.

The annealed sparsity reward is inert. Edit concentration stays in the 0.21–0.31 band in every arm. The sparsity term was built to address this and had been disabled throughout, leaving its annealing ramp inert. We enabled it and retrained. In n→p the validity safeguard (Supp. R1) froze $\lambda_{\text{sparsity}}$ at 0 before its ramp began, so only p→n trained with the term. There, concentration moved by less than 0.02, in-lung LHR did not change (paired $+0.026$, $[-0.003, +0.056]$, $p = 0.085$), and p→n transfer fell from 0.292 to 0.212 with non-overlapping CIs on two of three validators. Without a working spatial mask the term reduces to unweighted L_1, which shrinks all edit mass rather than off-target mass.

The Grad-CAM constraint collapses validity. Under the training-time Grad-CAM path, `chex`'s maps degenerate on 66 of 263 member-image observations (`rsna`'s on 1), at *low* predicted probability, consistent with output saturation zeroing the gradient. A finer target layer removes the degeneracy (0%/0.4%) but leaves CAM–lesion lift at 1.196, barely above chance. A policy trained under that remedy collapses CV from 0.808 to 0.0175 (7 of 400 valid) and lowers transfer on all three validators. Its in-lung LHR is not distinguishable from the default's (paired $+0.025$, $[-0.031, +0.080]$, $p = 0.37$). Its whole-image gap ($+0.960$, $d = 1.07$, the largest whole-image effect we measure) reflects edits leaving the lung (41% of edit mass in the lung vs. 66%), not worse targeting within it.

6. Limitations

- **One dataset, two generators.** Every result is on RSNA Pneumonia. Besides our system we ran Latent Shift and a CycleGAN under the identical protocol; DiME, LDCE and robust-classifier methods need RSNA-trained models that do not exist (Supp. E3). The CV gap is established for two systems, not for the field. An exploratory mammography port supports LHR in both directions, but its guidance classifier saw most evaluation images and no validator reaches AUROC 0.65 (Supp. E9). No radiologist assessed the counterfactuals, and we make no clinical claim.
- **Edits remain diffuse.** No arm we trained moved edit concentration meaningfully; every one sits in the same 0.21–0.31 band. The lock keeps at least 93% of the change relative to the VAE round trip inside the editable lung region (Supp. E6), so the diffuseness is within the lung. Whether diffuseness is fixable within a gradient-guidance approach is open.
- **Transfer anchors are approximate.** The ceiling compares populations, not paired edits, and the norm-matched attack is large enough to be partly semantic (§5.3). We therefore claim only that n→p transfer does not exceed the attack floor, not that the edits are adversarial.
- **Classifier overlap with the evaluation data.** The `rsna` guidance model and the FID extractor were trained on RSNA data, and `nih` on NIH ChestX-ray14, which contains RSNA's images (§4.3); these models may have seen our test images, which can inflate CV and the `nih` ratio. `chex`, `pc` and `mimic_ch` have no RSNA exposure. Generator fine-tuning, PPO rollouts and the validity probe use only the 80% training fold.
- **Few training runs per arm.** The default and single-classifier configurations have three training runs each, but only the PPO seed varies, except in one default run that also retrained the LoRA adapter, as one adapter for both directions; several runs resumed after preemption without saved random-number state (Supp. E10). The other arms have one run each, so their gaps have no run-to-run error bar. The single-classifier and ensemble runs also differ in classifier-selected groups (78 of 200 p→n sources shared) and hardware. Three arms were evaluated with the n→p adapter for both directions, which we match in their comparisons (§3.4); the Grad-CAM arm's n→p policy was trained for updates 1 to 49 under a different adapter from the one used at evaluation.
- **Adaptivity is limited.** In n→p, 199 of 200 episodes end after one step, with strength, pass count and guidance bias at or near fixed values; in p→n the median episode is one step and 67 to 78% of the action variance in multi-step episodes is between images (Supp. R3). A fixed schedule matches the policy (§5.2), so per-image adaptation adds nothing we can detect. The 37.9M-parameter policy was trained on at most 800×25 transitions per direction.
- **Sample size and proxies.** The 3-model arm's p→n group is small ($n = 32$, 27 annotated; 12 shared for LHR). FID at $n \leq 200$ in a 1024-d space is high-variance (Supp. E6). FOP is not computable offline, so IP measures a related property, and SCP is degenerate in this configuration.
- **Not measured:** convergence curves (two logged points per direction); a LoRA-rank ablation; Score-CAM or consensus masks; a wider a_7 range; n→p lesion targeting (LHR needs annotated sources; a second dataset with boxes, such as VinDr-CXR, would also anchor it). The lock width was not tuned (exploratory check in Supp. E4).

7. Conclusion

We evaluated a PPO-scheduled, LoRA-adapted latent diffusion counterfactual generator for chest radiographs under a protocol that pairs classifier-scored validity with an in-lung lesion hit rate and a held-out transfer ratio, each calibrated against explicit nulls. Under CV alone the system looks strong: 94.5% of normal-to-pneumonia edits flip the guidance classifier. The anchored metrics qualify that in three ways. In that direction, held-out classifiers move no more than they do under a norm-matched attack on the guidance models, and an iterative Latent Shift baseline behaves the same way. Pneumonia-to-normal edits do target lesions, but weakly (1.16× within the lung, half of it generic), and a fixed schedule at the policy's mean action does as well as the learned controller. And calibration matters as much as the metric: a whole-image null inflated lesion targeting sevenfold and produced arm rankings that vanish within the lung. Our evidence is two generators on one dataset, with few training runs per arm. Within it, CV computed with the guidance classifier overstates what the edits achieve, and anchored companion metrics are cheap, but informative only

once their own nulls are checked.

LLM Usage Statement

We used an LLM-based coding and writing assistant (Claude, Anthropic) extensively, under the authors' direction. (i) Code: it wrote much of the experiment and analysis code, including evaluation wrappers, the baselines' evaluation code, bootstrap and paired-test scripts, figure scripts and job-submission scripts. (ii) Experiments: in response to review feedback, it proposed, launched and analysed most of the follow-up experiments (the Latent Shift baseline, the LHR nulls, the transfer floor and ceiling, the seed replicates and the mammography LHR re-analysis). It also checked reported values against the logs and result files they came from. (iii) Writing: it drafted and revised large parts of the manuscript, tables and captions, and proposed framing changes, including the choice of in-lung LHR as the primary metric. The authors conceived the study and built the PPO-CF method and its original evaluation. They approved each experiment, framing decision and claim, and take full responsibility for the paper. No LLM was used to generate data, images or annotations, and no LLM is part of the method.

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